29/7/26.

Practical - 1 - Basic Network commands.

Aim: Study of various Network commands used in Linux & windows.

Basic Networking Commands :-

i) arp -a

purpose: It will show the IP address of your computer along with the IP address and MAC address of your router comand.

Command: arp -a

O/p : gateway (10.0.2.2) at 52.55:0a:00:02:02 [either] on enpos3.

ii) host name

purpose: Shows the computer name

command: hostname

O/p : localhost.localdomain.

iii) ipconfig :

purpose: it shows network configuration details

cmd: ipconfig

O/p : Wireless LAN adapter Wi-Fi :

Connection-specific DNS suffix :

Link-local IPv6 Address .... : Fe80::6768:d34:5781:ef2a v6

IPv4 Address . . . . . : 172.16.11.108

Subnet Mask . . . . . : 255.255.252.0

Default Gateway . . . . . : 172.16.8.1.

iv) Command: nslookup www.google.com

Description: Tool used to perform DNS lookups in Linux.

Output: Non-authoritative answer

Name: www.google.com.

Address: 142.251.152.119.

v) Command : pathping

Description : pathping is unique to windows and is basically a combination of ping and transient comds.

vi) Command : ping

Description : Test the connectivity between two nodes.

(i) ping local host

O/p :

64 bytes from local host (::1) : icmp - seq = 1

64 bytes from local host (::1) : icmp - seq = 2

ttl = 64 time = 0.125 ms.

(ii) ping 4.2.2.2

output :

64 bytes from 4.2.2.2 icmp - seq = 1 ttl = 52 time = 255 ms.

64 bytes from 4.2.2.2 icmp - seq = 2 ttl = 52 time = 221 ms

iii) ping www.Facebook.com

O/p :

64 bytes from edge - star - mini - shv - 02 - maas :

Facebook.com (57.144.56.1) : icmp - seq = 1

ttl = 1 time = 35.1 ms.

vii) Command : Route

Description : Its used to manipulate the IP routing table

output: kernel IP routing table

Destination	Gateway	Genmore	Flag	metric	Ref	use
default	- gateway	0.0.0.0	UH	100	0	0
10.0.2.0	0.0.0.0	255.255.255	U	100	0	0

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enp0s3
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Linux Networking Command.

1) Command: Ip

Description: It can show address information, manipulate various devices & tunnels.

Some Common use case Ip commands:

(i) Ip address show

output:

enp0s : <BROADCAST, multicast, UP, LOWER-UP>

Mtu 1500 qdisc fq_codel state up group default qlen 1000

link/ether 08:00:27:90:d3:d3

brd ff:ff:ff:ff:ff:ff

10.0.2.15/25 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3 valid_lft.

ii) ip address add 192.168.1.254/24 dev enp0s3

o/p:

inet 192.168.1.254/24 scope global enp0s3

The address has been added in enp0s3.

iii) ip address delete 192.168.1.254/25 dev enp0s3.

o/p: The ip address 192.168.1.254/25 was successfully deleted from the network interface enp0s3.

ii) Command: ifconfig

Description:- It is a state in many Sys admin's tool belt for configuring & troubleshooting networks.

Output:

lo: flags=73<up, LOOPBACK, RUNNING> MTU 65536

inet 127.0.0.1 network 255.0.0.0

inet ::1 prefixlen 128 scopeid 0x10<host>

loop txqueue.

iii) command : mtr

description : It serves as a network diagnostic & troubleshooting tool.

(i) mtr google.com

o/p :

(ii) mtr :-g google.com

o/p :

-u, -udp use upp instead of Icmp echo

-T, -tcp use Tcp instead of Icmp echo.

(iii) mtr -b google.com

o/p :-

Host	Loss	Snt	Last	Avg	Best	Worst
gateway	0.0%	9	0.8	1.4	0.5	5.5

1.3

iv) mtr -c 10

o/p :-

Host	Loss %	Snt	Last	Avg	Best	Worst
localhost	0.0%	2	0.1	0.1	0.1	0.4

Output :-

(i) dnf install -y tcpdump

o/p :

Dependies resolved

Nothing to do complete.

ii) tcpdump -D

o/p :-

bluetooth - monitor [wireless]

usbmon2 : [Raw USB traffic, bus number 2]

iii) tcp dump -i etho

o/p

iv) tcpdump -i etho -c 10

o/p

v) tcpdump -i etho

o/p

Result:

Thus the network command using linux & window are sucessfully executed & the o/p is verified.